

Homework Sample TeX Code

Economics 7103

1 Python

Variable	Control	Treatment	P-value
electricity	1181.33 (454.31)	1086.75 (423.96)	0.0007
sqft	1633.05 (682.90)	1657.55 (686.27)	0.5716
temp	79.89 (2.16)	79.89 (1.97)	0.9871
Observations	501	499	

Table 1: Sample mean table for control and treatment groups. Column 1 contains the variable names. Columns 2 & 3 show sample means for the control and treatment groups respectively. Column 4 shows the p-values for the difference-in-means test for each variable. Number of observations in each group are recorded at the bottom of the table. Standard errors are in parantheses.

It appears that randomization was successful. The treatment and control groups appear to be similar across all baseline characteristics (**sqft** and **temp**), excluding electricity usage. This implies that the difference-in-means estimate for the **electricity** variable can be attributed to the treatment effect, assuming no other confounding factors.

Variable	OLS by Hand	OLS by Simulation	OLS by StatsModels
Intercept	-83.602758	-83.605823	-83.602758
Sqft	0.615339	0.615338	0.615339
Retrofit	-109.666176	-109.666120	-109.666176
Temperature	3.255075	3.255114	3.255075

Table 2: OLS regression results. Dependent variable is monthly electricity usage. Column 1 contains the variable names. Column 2 shows the results for $\hat{\beta}$ using matrix operations by the formula $\hat{\beta} = (X'X)^{-1}X'Y$. Column 3 shows the results for $\hat{\beta}$ by minimizing the sum of squares objective function, i.e., $\sum_{i=1}^n (y_i - \beta \cdot x_i)^2$. Column 4 shows the results for $\hat{\beta}$ using the StatsModels package in Python.

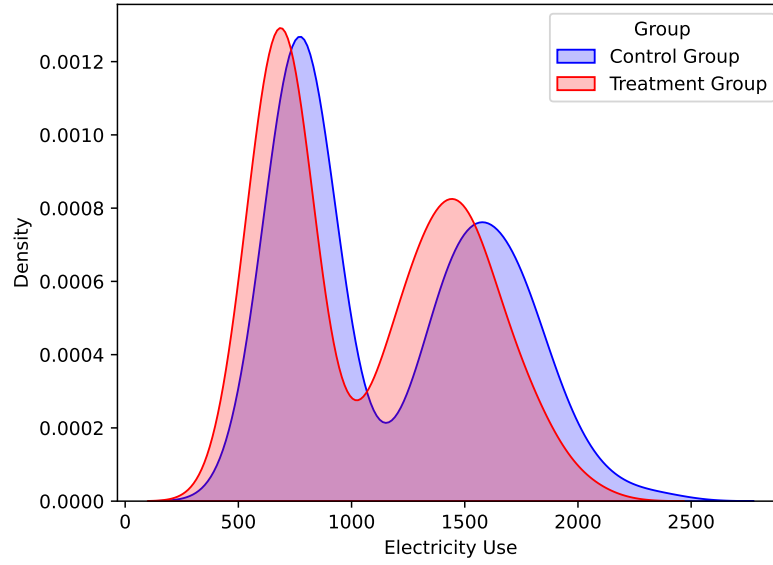


Figure 1: Kernel density plot of monthly electricity consumption for control and treatment groups.

2 STATA

	(1)	(2)	(3)
	Control	Treatment	Difference
electricity	1181.33 (454.31)	1086.75 (423.96)	94.584***
sqft	1633.05 (682.90)	1657.55 (686.27)	
temp	79.89 (2.16)	79.89 (1.97)	-24.499
			-0.002
Observations	501	499	1000

*** p<0.01, ** p<0.05

Table 3: Sample mean table for control and treatment groups. Columns (1) & (2) show sample means for the control and treatment groups respectively. Column (3) shows for the difference-in-means estimate for each variable. Number of observations are recorded at the bottom of the table. Standard errors are in parantheses.

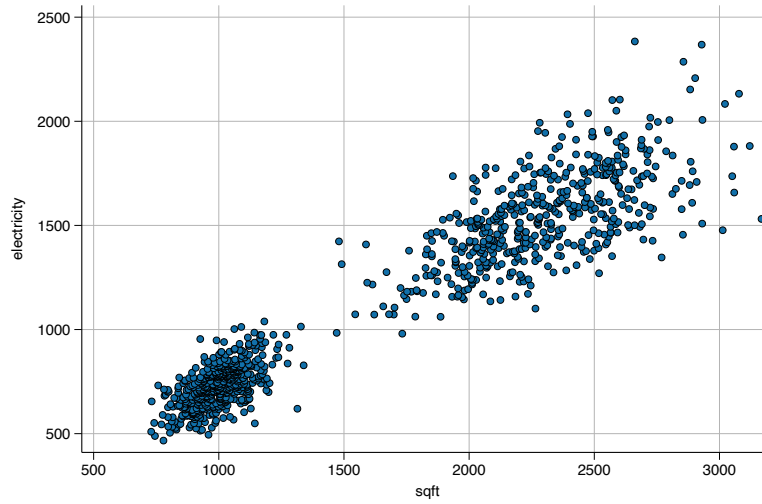


Figure 2: Scatterplot of monthly electricity usage against square feet of the the home.

(1)	
VARIABLES	Ordinary least squares
sqft	0.62** (0.01)
retrofit	-109.67** (7.94)
temp	3.26 (1.93)
Constant	-83.60 (154.69)
Observations	1,000
R-squared	0.92
Robust standard errors in parentheses	
** p<0.01, * p<0.05	

Table 4: OLS regression results. Dependent variable is monthly electricity usage. Column (1) shows coefficient estimates. Number of observations and R-squared are recorded at the bottom of the table.